

ASSEMBLY P/N	DESCRIPTION	REVISION	DOCUMENT	RELEASED	CHANGE REF
MC-4000-01	Motor Controller, 3-Phase, 48 V	B	BOM-MC-4000-01-B	2026-09-01	ECO-2026-0447
Prepared: R. Nair, Hardware Engineering Checked: S. Iyer, NPI Approved: M. Prasad, Engineering Manager Classification: Internal					

Item	Ref Des	Qty	Manufacturer	Manufacturer P/N	Description	Package	Lifecycle	RoHS	DNP	Approved Alternate
1	U1	1	STMicroelectronics	STM32F407VGT6	Cortex-M4F MCU, 168 MHz, 1 MB flash, L	LQFP-100	ACTIVE	Yes		STM32F429VGT6
2	U2	1	Texas Instruments	DRV8301DCAR	Three-phase gate driver with buck and	HTSSOP-56	ACTIVE	Yes		
3	Q1-Q6	6	Infineon	IRFB4110PBF	N-channel MOSFET 100 V 180 A, TO-220AB	TO-220AB	ACTIVE	Yes		
4	U3	1	Texas Instruments	TPS54331DDAR	3 A step-down converter, 3.5-28 V input	SOIC-8 Power	ACTIVE	Yes		
5	U4	1	Advanced Monolithic	AMS1117-3.3	1 A LDO, fixed 3.3 V, SOT-223	SOT-223	ACTIVE	Yes		NCP1117ST33T3G
6	U5	1	Texas Instruments	SN65HVD230DR	3.3 V CAN transceiver, 1 Mbps, SOIC-8	SOIC-8	ACTIVE	Yes		TCAN332DR
7	U6	1	Winbond	W25Q128JVS IQ	128 Mbit SPI NOR flash, SOIC-8 208 mi	SOIC-8 208mi	ACTIVE	Yes		MX25L12835FM2I-10G
8	Y1	1	Abracon	ABM8-25.000MHZ-D2Y-T	25 MHz crystal, 18 pF, 20 ppm, SMD 322	SMD-3225	ACTIVE	Yes		
9	D1	1	STMicroelectronics	USBLC6-2SC6	Dual-line USB ESD protection, SOT-23-6	SOT-23-6	ACTIVE	Yes		
10	C1-C32	32	Murata	GRM188R71H104KA93D	100 nF 50 V X7R 10% 0603	0603	ACTIVE	Yes		CL10B104KB8NNNC
11	C33-C42	10	Murata	GRM21BR61E106KA73L	10 uF 25 V X5R 10% 0805	0805	ACTIVE	Yes		
12	R1-R16	16	Yageo	RC0603FR-0710KL	10 kohm 1% 0.1 W 0603	0603	ACTIVE	Yes		
13	R17-R22	6	Yageo	RC0603FR-07100RL	100 ohm 1% 0.1 W 0603	0603	ACTIVE	Yes		
14	R23-R26	4	Yageo	RC0603FR-074K7L	4.7 kohm 1% 0.1 W 0603	0603	ACTIVE	Yes		
15	L1-L2	2	Bourns	SRN6045TA-100M	10 uH 3.0 A shielded power inductor, 6	SMD-6045	ACTIVE	Yes		
16	J1-J2	2	Molex	43045-0400	Micro-Fit 3.0 header, 4 position, right	Header	ACTIVE	Yes		
17	D2-D3	2	Littelfuse	SMAJ33A	400 W TVS diode, 33 V unidirectional,	DO-214AC	ACTIVE	Yes		
18	R27-R28	2	Yageo	RC0603FR-0710KL	10 kohm 1% 0.1 W 0603	0603	ACTIVE	Yes	DNP	
19	C43	1	Murata	GRM21BR61E106KA73L	10 uF 25 V X5R 10% 0805	0805	ACTIVE	Yes	DNP	
20	U7	1	Microchip	MCP2515-I/SO	Stand-alone CAN controller with SPI	SOIC-18	NRND	Yes		
21	U8	1	Texas Instruments	LM358BDR	Dual operational amplifier	SOIC-8	ACTIVE	Yes		

Notes

1. Lines marked DNP are shown on the assembly drawing and shall not be purchased or placed.
2. Approved alternates are drop-in equivalents released by engineering. Any other substitution requires an approved deviation.
3. All parts RoHS 3 (EU 2015/863) compliant. Certificates of conformance required at first article.
4. U1 and U6 are MSL 3: bake per J-STD-033 if floor life is exceeded. ESD sensitive, handle per ANSI/ESD S20.20.
5. U1 is programmed post-assembly with firmware MC4000-FW rev 2.4.
6. U7 is NRND. Engineering to confirm a second source before the next build release.
7. U8 LM358BDR: refer to the supplier datasheet issued with this release, Texas Instruments SLOS068AB Rev AB.
8. Total line items 21. Placements per assembly 90, excluding DNP.

Revision history

Rev	Date	Change ref	Description
A	2026-06-14	ECO-2026-0301	Initial release

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B	2026-09-01		ECO-2026-0447		Added CAN controller U7 and dual op-amp U8. Decoupling increased to 32 places. R27-R28 and C43 changed to DNP.					